EE7207 Neural Networks & Deep Learning

Self-Organizing Maps (SOM)

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Group: EE7207 Peer Tutoring

Valid until 3/20 and will update upon joining group

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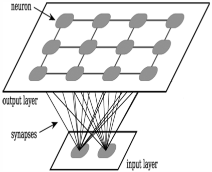
1. SOM: Architecture & Three Ingredients



- **Competition**: For each input, neurons calculate a discriminant function (usually Euclidean distance). The best match is the **BMU(Best Match Unit)**.
- **Cooperation**: The winning neuron defines a **topological neighborhood** of excited neurons in the lattice.
- **Adaptation**: Activated neurons adjust their weights to move closer to the input vector.

Past Year Quiz 1 Question

2. SOM: Training Process

1. **Initialization**: Assign small random values to weight vectors $w_j(0)$.
2. **Competitive Process**: Find BMU index $i(x) = \arg\min ||x - w_j||$.
3. **Adaptive Process**: Update weights of BMU and neighbors:

$$w_j(n+1) = w_j(n) + \eta(n) h_{ji(x)}(n)[x - w_j(n)]$$

4. **Continuation**: Repeat until convergence or stopping criterion.

Past Year Quiz 1 Question

3. Time-Varying Parameters

Crucially, SOM parameters must shrink over time n:

- **Learning Rate** $\eta(n)$: Starts high (~ 0.1) and decays to a small constant (~ 0.01).
- **Neighborhood Size** $\sigma(n)$: Initially covers almost all neurons, then shrinks to zero.
- **Decay Formula**:

$$\eta(n) = \eta_0 \exp\left(-\frac{n}{\tau_2}\right), \quad \sigma(n) = \sigma_0 \exp\left(-\frac{n}{\tau_1}\right)$$

4. Self-Organizing vs. Convergence Phases

The training is split into two distinct stages:

Feature	Self-Organizing Phase	Convergence Phase
Weights	Random values	From 1st phase
Learning Rate	Exponential decay	Small constant (0.01)
Neighborhood	Covers all/many	Only the BMU
Purpose	Topological ordering	Accurate quantification

5. Properties & Application of the SOM

After training, the SOM exhibits:

- **Approximation**: Small set of prototypes represents a large dataset.
- **Topological Ordering**: Spatial location in lattice corresponds to input domains.
- **Density Matching**: Map reflects input distribution statistics.
- **Dimensionality Reduction**: Maps high-dim data to a low-dim (2D) lattice.

Exam Case 2022-2023-S1-Q2(a)

2. A Bank has one million customers. In a direct marketing study, 2500 customers are to be selected as the representatives of the one million customers.

   (a) Sam suggested that the self-organizing map (SOM) neural network be used for the selection of the 2500 customers. Discuss the rationale of Sam's suggestion.

   (5 Marks)

   (b) Discuss how to design and train the SOM neural network for selecting the 2500 customers.

   (10 Marks)

   (c) SOM neural networks are often used for clustering. Discuss any other applications of SOM neural networks.

   (5 Marks)

- **Density Matching**: The map reflects the statistics of the input. Higher density regions in the 1M customer data will be represented by more neurons.
- **Vector Quantization**: 2,500 weight vectors act as "codebook vectors," providing an optimal **approximation** of the original dataset.
- **Topological Ordering**: Similar customers (e.g., similar income/spending) are mapped to adjacent neurons, making the 2,500 representatives structured.

- **Architecture**: A 2D lattice of $50 \times 50 = 2,500$ neurons. Input dimension m matches the number of customer attributes.
- **Initialization**: Weights w_j initialized with random values.

Training Steps:

1. **Competition**: Find BMU $i(x)$ for each customer x using $d_j(x) = \sum_{i=1}^{m}(x_i - w_{ji})^2$.
2. **Cooperation**: Determine neighborhood $h_{ji}(n)$, which shrinks over time.
3. **Adaptation**: Update weights: $\Delta w_j = \eta(n)h_{ji}(n)(x - w_j)$.
4. **Phases**:
    - *Ordering Phase*: Large η and σ for global topology.
    - *Convergence Phase*: Small η and σ for fine-tuning.

- **Connection to Clustering**: SOM performs **unsupervised clustering**. Each neuron (weight vector) represents a cluster center. After training, customers mapped to the same BMU belong to the same cluster.

Other Applications (5 Marks):

1. **Data Visualization (U-Matrix)**: Visualizing cluster boundaries in high-dim data.
2. **Dimensionality Reduction**: Like PCA, but non-linear. Mapping complex data to a 2D plane
3. **Vector Quantization**: Efficient data compression (e.g., image/speech coding).

Calculation Exercise

Scenario: A 2×2 lattice. Neuron 1 at $(0, 0)$ is BMU for input $x = [1, 0]$.

1. **Distance Calculation**: Lateral distance between BMU (1) and Neuron 4 at $(1, 1)$:

$$d_{14} = \sqrt{(1-0)^2 + (1-0)^2} = \sqrt{2} \approx 1.414$$

2. **Update Exercise**: If $\eta = 0.5$, $h = 0.6$, update w_4 given $w_{4,old} = [0, 0.5]$:

$$\Delta w_4 = 0.5 \cdot 0.6 \cdot ([1, 0] - [0, 0.5]) = [0.3, -0.15]$$

$$w_{4,new} = [0, 0.5] + [0.3, -0.15] = [\mathbf{0.3}, \mathbf{0.35}]$$

Summary: SOM Mind Map